

Hospital Readmission Annotation Instructions

Task: Annotate 200 hospital encounters as YES/NO for **30-day readmission risk**

Estimated Time: 15–30 minutes (~5 sec/row after understanding rules)

Format: CSV spreadsheet (Excel / Google Sheets)

1. Quick Reference Rule

- Count **strong risk factors** in each encounter (see Table below).
 - ≥ 2 strong factors $\rightarrow$ YES
 - 0–1 strong factors $\rightarrow$ NO
 - Borderline cases $\rightarrow$ consult Section 5 of `Annotation_Guidelines.pdf`
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2. Preparation

1. Review Guidelines:

Read `Annotation_Guidelines.pdf` for:

- Overview of your task
- Label definitions (YES / NO)
- Decision rules for the 7 strong factors
- 3 worked examples (YES, NO, and borderline)

2. Download Files:

- `annotation_template.csv` – your labeling spreadsheet
- `Annotation_Guidelines.pdf` – reference guide

3. Choose Software:

- Microsoft Excel, Google Sheets, Apple Numbers, or any CSV-compatible tool

Do not convert to .xlsx before annotating

3. Understanding the Spreadsheet

- **Rows:** 1 header + 200 encounters
- **Columns:** 46, including critical fields below

Column	Purpose	Strong Factor Rule
encounter_id	Unique ID	Reference only, do not edit
time_in_hospital	Length of stay	≥ 7 days = strong factor
number_inpatient	Prior admissions	≥ 2 = strong factor
number_diagnoses	Count of diagnoses	≥ 7 = strong factor
admission_type_id	1=Emergency	Emergency = strong factor
insulin	Med change	Up = strong factor
annotated_label	Your label	Type YES or NO

Other columns are informational. Only key columns above affect YES/NO decisions.

4. Annotation Process

1. Start with **Row 2**
2. Count strong factors using the key columns
3. Apply rule: $\geq 2 \rightarrow \text{YES}$, $0-1 \rightarrow \text{NO}$
4. Enter label in `annotated_label` column (exactly YES or NO)
5. Repeat for all 200 encounters

Example – Encounter 1:

Column	Value	Strong Factor?
time_in_hospital	11	✓
number_inpatient	0	✗
number_diagnoses	5	✗
admission_type_id	1	✓
insulin	Steady	✗

Strong factors = 2 → LABEL: YES

5. Quality Assurance

- Spot-check **every 50 rows**
 - Ensure ≥ 2 strong factors for YES, ≤ 1 for NO
 - Verify formatting: **YES / NO only** (no variations)
 - No blank cells in `annotated_label` column
 - Correct typos or extra spaces
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6. Saving & Submission

- **Excel:** File → Save As → `annotations_completed.csv` → CSV (Comma Delimited)
- **Google Sheets:** File → Download → CSV → rename to `annotations_completed.csv`
- Email the `annotations_completed.csv` to the researchers; emails are provided in section 8 (Contact & Support).

Before submitting:

- All 200 rows labeled
 - Formatting consistent: YES / NO
 - Filename: `annotations_completed.csv`
 - No typos or blank cells
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7. Troubleshooting Tips

- Cannot open CSV → Open with Excel/Sheets/Calc
- Spreadsheet won't save → Ensure CSV format selected
- Deleted a column → Re-download template, do not reorder columns
- Unsure about a row → Default to NO unless ≥ 2 strong factors

8. Contact & Support

Email: [alexgurung@umich.edu] [ckrai@umich.edu]

Include **Encounter ID** and brief question

9. Output Example

encounter_id	patient_nbr	race	gender	age	Annotated_label
110939484	19274094	Caucasian	Female	[70-80]	YES
170328306	65634327	Caucasian	Male	[50-60]	NO
245688426	100657359	Caucasian	Female	[60-70]	NO

[rows 4-200...]

Thank you! Your annotations support research, validate data quality, and contribute to building interpretable ML tools for hospital readmission prediction.